$$= 1-\frac{132}{990} = 1-0.1333=0.8667$$

$$\rho_{ZX} = 1-\frac{6\Sigma d_{ZX}^2}{N(N^2-1)} = 1-\frac{6(158)}{10(10^2-1)}$$

$$= 1-0.9576 = 0.0424$$

Since the rank correlation coefficient between Second and Third judges i.e., ρ_{YZ} is positive and weight among the three coefficients. So, Second judge and Third judge have the nearest approach for common taste in beauty.

Example 9.8

Calculate rank correlation coefficient of the following data

Subject 1	40	46	54	60	70	80	82	85	87	90	95
Subject2	45	46	50	43	40	75	55	72	65	42	70

Solution:

Let X is considered for Subject 1 and

Y is considered for Subject 2.

X	Y	R_X	R_Y	$d = R_X - R_Y$	d^2
40	45	1	4	–3	9
46	46	2	5	–3	9
54	50	3	6	–3	9
60	43	4	3	1	1
70	40	5	1	4	16
80	75	6	11	–5	25
82	55	7	7	0	0
85	72	8	10	–2	4
87	65	9	8	1	1
90	42	10	2	8	64
95	70	11	9	2	4
					$\Sigma d^2 = 142$

Table 9.6

$$\rho = 1-\frac{6\Sigma d^2}{N(N^2-1)}$$

$$= 1-\frac{6(142)}{11(11^2-1)}$$

$$= 1-\frac{852}{1320} = 0.354$$

Exercise 9.1

1. Calculate the correlation co-efficient for the following data

X	5	10	5	11	12	4	3	2	7	1
Y	1	6	2	8	5	1	4	6	5	2

2. Find coefficient of correlation for the following:

Cost (₹)	14	19	24	21	26	22	15	20	19
Sales (₹)	31	36	48	37	50	45	33	41	39

3. Calculate coefficient of correlation for the ages of husbands and their respective wives:

Age of husbands	23	27	28	29	30	31	33	35	36	39
Age of wives	18	22	23	24	25	26	28	29	30	32

4. Calculate the coefficient of correlation between X and Y series from the following data

Description	X	Y
Number of pairs of observation	15	15
Arithmetic mean	25	18
Standard deviation	3.01	3.03
Sum of squares of deviation from the arithmetic mean	136	138

Summation of product deviations of X and Y series from their respective arithmetic means is 122.

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